navigation
- Main page
- Community portal
- Current events
- Recent changes
- Random page
- Help
- All pages

search
Search Whole Sky
Go Search

tools
- What links here
- Related changes
- Special pages
- Printable version
- Permanent link
- Page information

WSDB Euclid Q1

Euclid Q1 data

I store the Euclid merged catalogue:

euclid_q1.mer_final_cat

See columns:

[redacted]https://euclid.esac.esa.int/dr/q1/dpdd/merdpd/dpcards/mer_finalcatalog.html

See for the guidance: [redacted]https://euclid.esac.esa.int/dr/q1/dpdd/merdpd/merphotometrycookbook.html

Sources were downloaded from [redacted]https://irsa.ipac.caltech.edu/data/Euclid/docs/overview_q1.html

The number of sources: 29953430 For some reason this paper [redacted]https://arxiv.org/html/2503.15305v2#S8 lists 29767806 this many sources

The indices were created on right ascension, declination and object_id

Here's ChatGPT's distillation of columns:

Contents [hide]

1 1. Identification & Coordinates
2 2. Detection & Basic Properties
3 3. Flux Measurements: Aperture & Errors
4 4. Flux Measurements: Template-Fit (TPhot)
5 5. Flux Measurements: PSF, Segmentation, & Detection Totals
6 6. Flux Measurements: Sersic & Disk-Sersic
7 7. Sersic Fract (Fraction) Columns
8 8. Flags per Band
9 9. Photometric Calibration & Filter Info
10 10. Additional Quality & Classification
11 11. Morphological & Geometrical Parameters
12 12. Galactic Extinction
13 13. Gaia Cross-match

1. Identification & Coordinates [edit]

- **object_id**
- **right_ascension / declination**
- **right_ascension_psf_fitting / declination_psf_fitting**
- **segmentation_map_id**

2. Detection & Basic Properties [edit]

- **vis_det** (indicates if source detected in VIS or only in NIR)
- **deblended_flag**, **parent_id**, **parent_visnir**, **blended_prob**
- **she_flag**, **variable_flag**, **binary_flag**
- **point_like_flag**, **point_like_prob**
- **extended_flag**, **extended_prob**
- **spurious_flag**, **spurious_prob**

3. Flux Measurements: Aperture & Errors [edit]

Here we have multiple columns repeated for each band (VIS, Y, J, H, NIR stack, etc.) and for each external band (u/g/r/i/z from DECam, LSST, Megacam, JPCam, Pan-STARRS, HSC). Typically there are 4 aperture sizes:

- For example: **flux_[band]_[1..4]fwhm_aper** and **fluxerr_[band]_[1..4]fwhm_aper**
- This pattern applies to each band: VIS, Y, J, H, NIR_stack, U_EXT_DECAM, G_EXT_DECAM, R_EXT_DECAM, etc.

4. Flux Measurements: Template-Fit (TPhot) [edit]

Similarly repeated for each band but labeled **_TEMPLFIT** rather than **_APER**. These also have corresponding errors:

- For example: **flux_[band]_templfit**, **fluxerr_[band]_templfit**
- Additionally, there are columns with VIS matched to other bands: **flux_vis_to_[band]_templfit** and **fluxerr_vis_to_[band]_templfit**

5. Flux Measurements: PSF, Segmentation, & Detection Totals [edit]

- **flux_vis_psf**, **fluxerr_vis_psf**
- **flux_segmentation**, **fluxerr_segmentation**
- **flux_detection_total**, **fluxerr_detection_total**

6. Flux Measurements: Sersic & Disk-Sersic [edit]

Each band can have a Sersic fit flux (and error) or a Disk-Sersic fit flux (and error):

- For example: **flux_[band]_sersic**, **fluxerr_[band]_sersic**
- For Disk-Sersic: **flux_[band]_disk_sersic**, **fluxerr_[band]_disk_sersic**

7. Sersic Fract (Fraction) Columns [edit]

Each band has columns indicating the fraction of flux modeled by the disk vs. bulge (Sersic):

- For example: **sersic_fract_[band]_disk_sersic** and **sersic_fract_[band]_disk_sersic_err**

8. Flags per Band [edit]

Flags that track pixel issues or other quality flags in each band:

- E.g., **flag_vis**, **flag_y**, **flag_j**, **flag_h**, **flag_nir_stack**
- And for external bands: **flag_u_ext_decam**, **flag_g_ext_decam**, etc.

9. Photometric Calibration & Filter Info [edit]

Average filter transmission wavelengths for each band:

- For example: **avg_trans_wave_vis**, **avg_trans_wave_y**, etc.
- And similarly for external bands: **avg_trans_wave_u_ext_decam**, etc.

10. Additional Quality & Classification [edit]

- **det_quality_flag**
- **mag_stargal_sep**
- **mu_max**, **mumax_minus_mag**
- **she_flag** (e.g., bright-star removal)
- (Many of these also appear in section 2, but you can group them here or keep them separate.)

11. Morphological & Geometrical Parameters [edit]

- **segmentation_area**
- **semimajor_axis**, **semimajor_axis_err**
- **position_angle**, **position_angle_err**
- **ellipticity**, **ellipticity_err**
- **kron_radius**, **kron_radius_err**
- **fwhm**

12. Galactic Extinction [edit]

- **gal_ebv**, **gal_ebv_err**

13. Gaia Cross-match [edit]

- **gaia_id**
- **gaia_match_quality**